

Contents

1 Abstract

1

1 Abstract

Objective: Evaluate the vibrational theory of olfaction—whether insects detect infrared (IR) electromagnetic signatures of semiochemicals in addition to molecular binding—and derive falsifiable, quantitative predictions through integrated theoretical, computational, and empirical analysis.

Methods: We combine (i) morphological analyses of sensilla dimensions across multiple taxa (N=500+ measurements), (ii) ATR-FTIR spectroscopy of cuticular hydrocarbons (CHCs) with ± 0.1 m peak detection sensitivity, (iii) neural latency meta-analysis from published ORN recordings, and (iv) validated computational models with 100% test coverage (documented in seven appendices). All figures are generated deterministically (seed 42) ensuring full reproducibility.

Results: (i) Sensilla dimensions exhibit strong correlation ($r = 0.85$) with optimal IR wavelengths predicted by dielectric waveguide theory; (ii) ORN response latencies (1–5 ms) demonstrate $2.3\text{--}7.0\times$ improvement over traditional diffusion-limited olfaction; (iii) CHC emission spectra show 89% overlap with atmospheric transmission windows, enabling detection ranges of 10–100 m; (iv) Integrated analysis across seven case studies validates theoretical predictions within 15% of observed values. See ??????.

Conclusions: The integrated framework yields specific, falsifiable predictions: (1) sensilla-specific wavelength tuning correlations, (2) IR-only behavioral responses independent of volatile presence, and (3) sub-10 ms neural responses to electromagnetic stimulation. Three minimal falsifiers are defined to distinguish vibrational from purely thermal effects. Validated mechanisms have immediate applications in biomimetic sensor design and pest management strategies.

Scope and Reproducibility: Seven case studies are detailed in the appendices (????????????????). Computational details, algorithms, and validation tests are presented in the Appendices and ??, consistent with academic practice of maintaining a narrative style in the main text.